

## Homework 2

**Due date: 05/10/2025**

### On Brownian motion

Consider a Brownian particle in the overdamped regime. The equation of motion in one-dimensional space is given by

$$\gamma \frac{dx}{dt} = -\frac{d}{dx}U(x) + \zeta ,$$

in which  $\gamma$  is the damping coefficient,  $U(x)$  is the external potential, and  $\zeta$  is the white noise satisfying the autocorrelation  $\langle \zeta(t_1)\zeta(t_2) \rangle = 2\gamma k_B T \delta(t_1 - t_2)$ . For simplicity, let's take  $\gamma = 1$  and  $k_B T = 1$  in the numerical simulations.

(a) For  $U(x) = 0$ , the Brownian particle undergoes a random walk. Please carry out the simulation to obtain the particle trajectory and compute  $\langle [x(t) - x(0)]^2 \rangle$ , which is expected to be linear in  $t$ . Determine the diffusivity from the numerical data for  $\langle [x(t) - x(0)]^2 \rangle$  and compare it with the theoretical value.

(b) For  $U(x) = x^2 / 2$  (a confinement potential), please carry out the simulation to find the stationary probability density function (PDF)  $f(x)$  from long-time behaviour of particle trajectory. Is your numerical PDF close to a normal distribution? What is the value of the variance of  $f(x)$ ?

Note that you can use any computer language to simulate the stochastic processes.

**Please submit both the computer program and the plots.**